

Singular Homology

1.1 The Construction

Definition 1.1.1

Let X be any topological space and $n \geq 0$ be an integer. By a **singular n -simplex** in X , we mean a continuous map $\sigma : |\Delta_n| \rightarrow X$. By a **singular n -chain** in X , we mean a finite sum $\sum_i n_i \sigma_i$, where $n_i \in \mathbb{Z}$ and σ_i are singular n -simplices in X . We can add any two chains in an obvious way to obtain another chain by simply following the rule

$$n\sigma + m\sigma = (n + m)\sigma \quad (4.5)$$

Then the collection of all singular n -chains in X becomes a free abelian group, with the empty sum playing the role of the zero-element. The collection of all singular n -simplices is then a basis for this group. We denote this group by $S_n(X)$.

Proposition 1.1.2 ► Functoriality

If $f : X \rightarrow Y$ is a map of topological spaces, then $\sigma \mapsto f \circ \sigma$ extends to define a graded group homomorphism $f_* : S_*(X) \rightarrow S_*(Y)$. Since $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$, it follows that the association $X \mapsto S_*(X)$ defines a functor.

Definition 1.1.3 ► Relative Chain Complex

If A is a subspace of X and σ is a singular simplex in A then it can be considered as a singular simplex in X via the inclusion map $A \hookrightarrow X$. In this way, $S_n(A)$ becomes a subgroup of $S_n(X)$. We denote the quotient group $S_n(X)/S_n(A)$ by $S_n(X, A)$.

Remark.

1. $S_n(X, A)$ is a free abelian group with the collection of all singular n -simplexes σ in X whose image is not contained in A as a basis.
2. If $A = \emptyset$, then clearly $S_n(X, A) = S_n(X)$.

Definition 1.1.4

For each integer $r \geq 0$, the *face map* or the *face operator* $F^r : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ is defined by

$$F^r(\mathbf{e}_s) = \begin{cases} \mathbf{e}_s, & \text{if } s < r, \\ \mathbf{e}_{s+1}, & \text{if } s \geq r \end{cases}$$

and extended linearly.

Remark.

1. Note that for each $n \geq r$, F^r carries Δ_{n-1} onto the $(n-1)$ -face of Δ_n opposite to the vertex $\mathbf{e}_r$, in an *order preserving* manner.
2. We shall denote each F^r restricted to any Δ_n , $n \geq r$ also by F^r , the exact meaning being understood from the context.

Lemma 1.1.5

$$F^r \circ F^{s-1} = F^s \circ F^r \text{ if } r < s. \text{ }^a$$

^a这里修正了 shastri 书中的一个笔误, 他可能写错了 $s-1$ 的位置

Proof. Since $r < s$, $r \leq s-1$, we fully expand the LHS by definition

$$\begin{aligned} F^r \circ F^{s-1}(\mathbf{e}_k) &= \begin{cases} F^r(\mathbf{e}_k), & k < s-1 \\ F^r(\mathbf{e}_{k+1}), & k \geq s-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < s-1, k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+1}, & s-1 \leq k < r-1 \text{ (empty set)} \\ \mathbf{e}_{k+2}, & k \geq s-1, k \geq r-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases} \end{aligned}$$

Similarly,

$$\begin{aligned}
 F^s \circ F^r (\mathbf{e}_k) &= \begin{cases} F^s (\mathbf{e}_k), & k < r \\ F^s (\mathbf{e}_{k+1}), & k \geq r \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r, k \leq s \\ \mathbf{e}_{k+1}, & k < r, k \geq s \text{ (empty set)} \\ \mathbf{e}_{k+1}, & k \geq r, k < s-1 \\ \mathbf{e}_{k+2}, & k \geq r, k \geq s-1 \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases}
 \end{aligned}$$

□

Definition 1.1.6

For $n \geq 1$ and for any singular n -simplex σ in X , we define

$$\partial_n(\sigma) = \sum_{r=0}^n (-1)^r \sigma \circ F^r$$

and extend it linearly to obtain homomorphisms $\partial_n : S_n(X, A) \rightarrow S_{n-1}(X, A)$. We define $S_n(X, A) = (0)$ for $n \leq -1$ and $\partial_n = 0$ for $n \geq 0$.

Note.

σ is a map from Δ_n to X . $\sigma \circ F^r$ is the restriction of σ on its face opposite to $\mathbf{e}_r$. Then $\partial_n(\sigma)$ is something like a closed $(n-1)$ -surface consists of σ 's faces.

Proposition 1.1.7 ► Singular Chain Complex

$(S(X, A), \partial) = \{S_n(X, A), \partial_n\}$ is a chain complex and the association $(X, A) \rightsquigarrow (S(X, A), \partial)$ is a functor.

Proof. We shall first show that $\partial^2(\sigma) = 0$ for any singular n -simplex, $n \geq 2$:

$$\begin{aligned}
 \partial^2(\sigma) &= \partial \left(\sum_{s=0}^n (-1)^s \sigma \circ F^s \right) \\
 &= \sum_{s=0}^n (-1)^s \partial(\sigma \circ F^s) \\
 &= \sum_{s=0}^n \sum_{r=0}^{n-1} (-1)^s (-1)^r \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq r \geq s \geq 0} (-1)^{s+r} \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq s \geq r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^s \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} - \sum_{n \geq s > r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} \\
 &= 0
 \end{aligned}$$

Given a map $f : (X, A) \rightarrow (Y, B)$. Since $(f \circ \sigma) \circ F^r = f \circ (\sigma \circ F^r)$, we define $f_*(\sigma) := f \circ \sigma$ extended by linearity. Then $\partial(f_*(\sigma)) = f_*(\partial(\sigma))$ by the definition of ∂ and f_* 's linearity. The rest of the details are omitted. $\square$

Definition 1.1.8

The **relative singular homology groups** of a topological pair (X, A) are the homology groups $H_*(S_*(X, A))$ and are denoted by $H_*(X, A)$. Taking $A = \emptyset$, we get the homology groups $H_*(X)$.

Remark.

Given any commutative ring R

1. We could have formed the **free module** $S_n(X; R)$ of all the singular n -chains in X .
2. Exactly as before, we would have got a chain complex $S_*(X; R)$, sub-chain complex $S_*(A; R)$ and the quotient complex $S_*(X, A; R)$, etc.
3. The corresponding homology groups $H_*(X; R)$, etc., are then called the homology groups X with coefficients in R . Clearly they form a **graded R -module** by themselves. It is customary to drop the notation R when $R = \mathbb{Z}$.

1.2 Basic Properties

Remark.

- It is easily seen that $\{S_*(X, A), \partial\}$ is a functor from the category of topological pairs to the category of chain complexes; thus $H_*(X, A)$ and $H_*(X)$ are also functors.
- Given $f : (X, A) \rightarrow (Y, B)$, we shall denote by f_* , the map induced by f on $S_*(X, A)$ and by f_* the map induced by f on $H_*(X, A)$.

Lemma 1.2.1

There exist functorial chain homotopies

$$h : S_q(X, A) \rightarrow S_{q+1}((X, A) \times \mathbb{I}), q \geq 0,$$

,called **prism operator**, between the two inclusion induced maps $\eta_0, \eta_1 : S_*(X, A) \rightarrow S_*((X, A) \times \mathbb{I})$, where for any $t \in \mathbb{I}$, η_t is the inclusion map

$$\eta_t(x) = (x, t), \quad t \in \mathbb{I}$$

Theorem 1.2.2 ▶ Homotopy invariance

Let $f, g : (X, A) \rightarrow (Y, B)$ be **homotopic** to **each other**. Then $f_* = g_*$ on $H_*(X, A)$.

Proof. Let $G : (X, A) \times \mathbb{I} \rightarrow (Y, B)$ be a homotopy between f and g . For any $t \in \mathbb{I}$, let $\eta_t : (X, A) \rightarrow (X, A) \times \mathbb{I}$ be inclusion map $\eta_t(x) = (x, t)$. Then $G \circ \eta_0 = f$, $G \circ \eta_1 = g$. Only need to show that chain maps $(\eta_0)_*$ and $(\eta_1)_*$ are chain homotopic, thus $(\eta_0)_* = (\eta_1)_*$, $f_* = g_*$, which is implied by the lemma above. $\square$

Corollary 1.2.3

Homotopy-equivalent topological spaces have isomorphic homology groups; in particular, a contractible space has the homology groups of a point-space.

Example 1.2.4

Let $\star$ be a point space. We have

$$H_k(\star) = \begin{cases} \mathbb{Z}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

Proof. Let n be any positive integer. There is a unique singular n -simplex $\sigma : |\Delta_n| \rightarrow \{\star\}$ and hence $S_n(\star) \cong \mathbb{Z}$ for each $n \geq 0$. Observe that $\partial : S_{2n+1}(\star) \rightarrow S_{2n}(\star)$ is the zero-map because Δ_{2n+1} has an even number of $(2n)$ -faces and the terms in $\partial(\sigma)$ cancel in pairs. However, similar reasoning tells us that $\partial : S_{2n}(\star) \rightarrow S_{2n-1}(\star)$ is an isomorphism. Thus we have a chain complex

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0$$

Thus it follows that $H_0(\star) \cong \mathbb{Z}$ and $H_i(\star) = (0)$ for all $i > 0$. $\square$

Example 1.2.5

Let $\{X_j : j \in J\}$ be the set of path connected components of X so that $X = \bigsqcup_{j \in J} X_j$. Then

$$H_*(X) = \bigoplus_{j \in J} H_*(X_j)$$

Remark.

Thus in practice, we can often assume that the space itself is path connected, while dealing with its homology groups.

Proof. Since $|\Delta_n|$ is a path connected space, it follows that any singular n -simplex in X is contained completely in one of the path components X_j . Therefore, it follows that the entire chain complex $S_*(X)$ is the direct sum of the subchain complexes $S_*(X_j)$. Hence the homology is also a direct sum of $H_*(X_j)$, $j \in J$ by Theorem ?? $\square$

Theorem 1.2.6 ▶ Homology Long Exact Sequence of the pair (X, A)

By definition of $S_*(X, A)$, we have an exact sequence of chain complexes

$$0 \rightarrow S_*(A) \rightarrow S_*(X) \rightarrow S_*(X, A) \rightarrow 0.$$

By Theorem ??, this yields the long exact sequence of homology groups

$$\begin{cases} \dots \rightarrow H_i(A) \rightarrow H_i(X) \rightarrow H_i(X, A) \xrightarrow{\delta} H_{i-1}(A) \rightarrow \dots \\ \dots \rightarrow H_1(X, A) \xrightarrow{\delta} H_0(A) \rightarrow H_0(X) \rightarrow H_0(X, A), \end{cases} \quad (4.6)$$

which is functorial.

Remark.

This sequence comes handy in many computations as we shall see soon.